

Project Report

Machine Learning (CS3102)

Hybrid Predictive Maintenance using Enhanced CMAPSS NASA Dataset

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CCP: Cluster, Classify, Predict

Abstract

Traditional predictive maintenance models often reduce complex mechanical degradation to a binary failure prediction, missing the gradual progression that real-world systems like aircraft engines undergo. This lack of granularity can delay actionable insights and lead to unscheduled downtimes or unsafe operations. In this paper, we propose a hybrid predictive maintenance system using the enhanced NASA CMAPSS dataset to model multi-stage degradation. We first employed unsupervised clustering on sensor readings to define five degradation stages, ranging from ‘Normal’ to ‘Failure’. A supervised classification model was then trained to identify the current health stage of the engine. In parallel, a regression model was developed to predict the remaining time until the next degradation stage. Finally, we introduced a composite Risk Score that integrates both the failure probability and estimated time-to-failure, enabling proactive and prioritized maintenance alerts. This multi-model approach bridges the gap between theoretical machine learning and practical industrial reliability requirements.

1 Introduction

Predictive maintenance has become a vital requirement in industries that rely on complex machinery, such as aviation, manufacturing, and energy. Traditional predictive models often simplify maintenance needs into binary outcomes—either a machine is working or it is about to fail. This oversimplification ignores the gradual degradation that systems typically undergo, resulting in missed opportunities to plan maintenance actions during early warning stages.

Such binary systems fall short when applied to real-world scenarios, where degradation is not abrupt but evolves over multiple stages—from optimal performance to eventual failure. This gap creates challenges in timely intervention, leading to either over-maintenance or unexpected breakdowns.

To address this issue, we developed a flexible hybrid pipeline that combines unsupervised and supervised learning techniques. The process begins with unsupervised clustering to automatically generate meaningful labels for degradation stages from raw sensor data. This stage-creation method is adaptable and can be applied to any dataset where standard labels may be absent or insufficient. Once the clusters representing different degradation levels are defined, the next step involves training a supervised classification model that can accurately predict the current degradation stage of a machine based on live sensor input.

To demonstrate the viability of our pipeline, we utilise the NASA CMAPSS dataset [12]. The NASA C-MAPSS (Commercial Modular Aero-Propulsion System Simulation) dataset provides simulated multi-sensor time-series readings (e.g., temperatures, pres-

tures, speeds) from many turbofan engines throughout their operational life until failure. It comprises four subsets (FD001–FD004) with varying fault modes and operating conditions, and serves as a standard benchmark for Remaining Useful Life (RUL) prediction and prognostics research.

We began with feature engineering on the CMAPSS dataset, followed by a weighted K-Means clustering approach [9], where lifecycle features were assigned higher weight to improve stage distinction. The clustered data was used to train a Random Forest classifier[1], which demonstrated high performance as evidenced by evaluation metrics and a clear confusion matrix. We then applied a Random Forest Regressor to predict Remaining Useful Life (RUL) until the next degradation stage, using metrics such as RMSE, MAE, and R^2 Score to assess performance. Finally, we introduced a composite Risk Score to support real-time decision-making. This score combines the classifier’s probability of failure with the regression-based estimate of time left until that failure. If the normalized risk score crosses a critical threshold, the system issues a proactive maintenance alert, thus enabling timely intervention and optimizing machine health management.

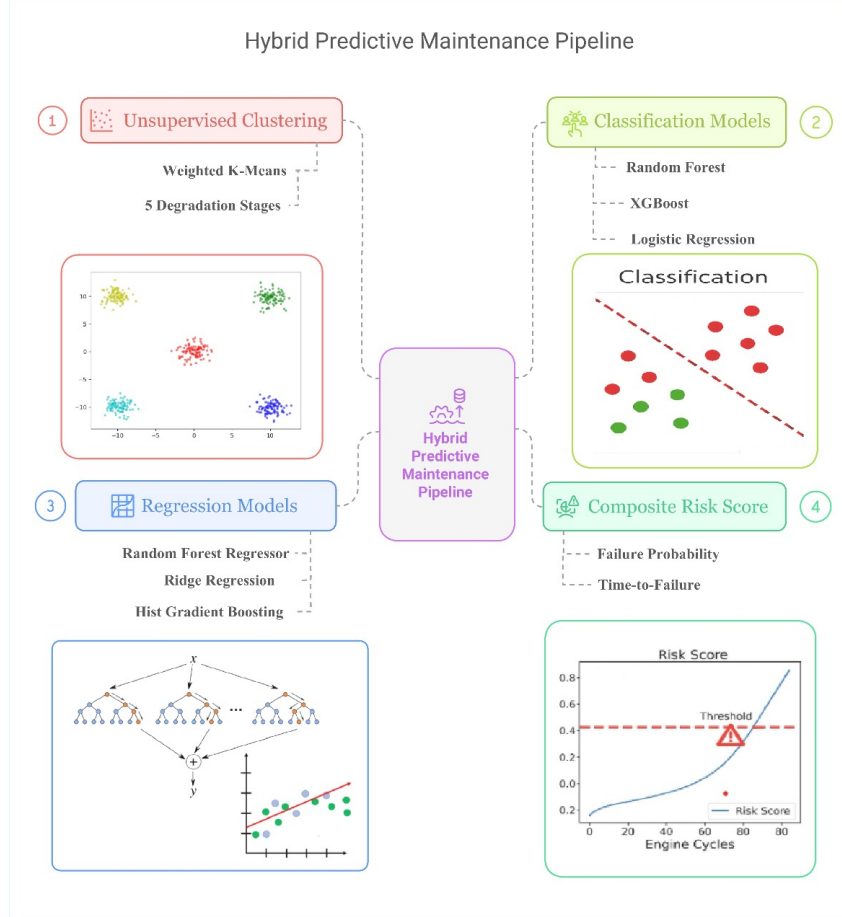


Figure 1: Hybrid predictive-maintenance workflow. (1) **Unsupervised clustering** applies weighted k -means to segment engine behaviour into *five degradation stages*. (2) **Classification** models assign new observations to those stages. (3) **Regression** models project *stage-wise remaining useful life*. (4) **Composite risk score** fuses stage and RUL estimates into failure-probability / time-to-failure curves for maintenance decision-making.

2 Dataset

The NASA CMAPSS (Commercial Modular Aero-Propulsion System Simulation) turbofan engine dataset is a widely recognized benchmark used in predictive maintenance and Remaining Useful Life (RUL) estimation. It simulates the behavior of turbofan engines under varying operational conditions and fault scenarios, capturing degradation patterns through multivariate time-series data.

The dataset is split into four subsets: FD001, FD002, FD003, and FD004. Each subset represents different combinations of operating conditions and fault modes, enabling the study of model generalization across diverse scenarios. Every engine unit is monitored over multiple operational cycles until failure, with each cycle containing measurements of three operational settings and 21 sensor readings such as temperatures, pressures, and vibration indicators.

The dataset’s realistic engine simulation and complexity makes it an ideal foundation for research in hybrid predictive maintenance frameworks that integrate classification, regression, and risk-based decision logic.

Table 1: Overview of CMAPSS Dataset Variants

Dataset	Operating Conditions	Fault Modes	Training Samples
FD001	1 (Sea Level)	HPC Degradation	20,631
FD002	6	HPC Degradation	53,070
FD003	1 (Sea Level)	HPC and Fan Degradation	24,710
FD004	6	HPC and Fan Degradation	61,984

3 Feature Selection

We performed basic feature engineering to prepare the dataset for model training. Initially, we carried out feature selection by removing columns that showed constant values, high missing rates, or low variance, as they did not contribute meaningfully to the learning process. The remaining numerical features were standardized by subtracting the mean and dividing by the standard deviation. This ensured that all features were on a similar scale, which is important for many machine learning algorithms, especially those sensitive to feature magnitudes. We did not apply any advanced feature extraction or domain-specific transformations, keeping the data close to its original form for interpretability and reproducibility.

4 Methodology

4.1 Phase 1: Clustering and Label Generation

The first challenge we encountered in our pipeline was the lack of explicit labels in the CMAPSS dataset. While the dataset provides time-series sensor data for multiple en-

gines, it does not contain any predefined categories or annotations indicating the health condition or degradation stage of each engine cycle. Since our downstream tasks required stage-wise classification and Remaining Useful Life (RUL) prediction, we needed a way to segment the operational timeline of engines into meaningful degradation stages. Given this unsupervised nature of the data, clustering was the most viable option to derive pseudo-labels that represent different health states. We aimed to identify five distinct degradation stages ranging from early (healthy) to late (near-failure) phases.

To achieve this, we first explored several well-known unsupervised clustering algorithms, including K-Means, Agglomerative Clustering[7], Gaussian Mixture Model [5], and HDBSCAN [2]. These models were applied using a selected subset of relevant features, primarily composed of sensor readings and operational settings. Importantly, we applied clustering separately on each of the four FD sub-datasets (FD001 to FD004), acknowledging that each dataset corresponds to a different combination of engine type, operational profile, and fault mode. This per-dataset approach ensured that clustering results were tailored to the specific dynamics of each engine group and preserved their internal consistency. Moreover, we also performed clustering on the combined FD001 $\cup$ FD003 and FD002 $\cup$ FD004 datasets, applying all clustering algorithms (K-Means, Agglomerative Clustering, GMM, HDBSCAN) to both single and merged dataset configurations. We additionally applied all clustering algorithms to the fully combined dataset (FD001 $\cup$ FD002 $\cup$ FD003 $\cup$ FD004) to capture global degradation patterns across the entire engine fleet.

We begin from the premise that degradation is proportional to operating cycles: as the cumulative number of cycles increases, the engine’s degradation accelerates.

However, when we applied off-the-shelf clustering algorithms-such as vanilla K-Means and Gaussian Mixture Models- the resulting groups often failed to align with the true progression of wear. In many instances, these methods produced clusters with weak or no correspondence to the gradual-to-rapid degradation pattern observed in practice. This shortcoming stems from their inherent assumption that all features contribute equally, thereby overlooking the critical, time-dependent nature of engine aging.

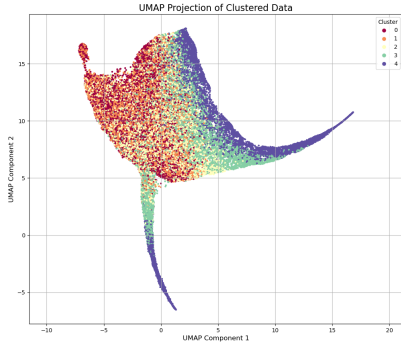
To overcome this, we used a weighted K-Means clustering approach, guided by the assumption that degradation accelerates with engine age, i.e., cycles later in the operational timeline are more reflective of impending failure than earlier ones. We operationalized this assumption by assigning higher weight to the cycle feature, making the algorithm more sensitive to lifecycle progression during clustering. The cycle feature was multiplied by weight w , while all other columns were normalized to be in $[-1, 1]$. This ensured that the clusters were influenced not just by instantaneous sensor readings, but also by where in the engine’s life those readings occurred.

This weighted approach yielded significantly more interpretable and well-separated clusters, with clearer differentiation between early, mid, and late degradation stages. Once these optimal clusters were obtained, we used the resulting cluster assignments as stage labels, thereby converting the originally unsupervised data into a structured multi-class classification problem. To further validate and visualize the effectiveness of our clustering, we applied Principal Component Analysis (PCA) [11] and UMAP [10] to reduce the feature space. The projections revealed well-separated clusters in the reduced-dimensional

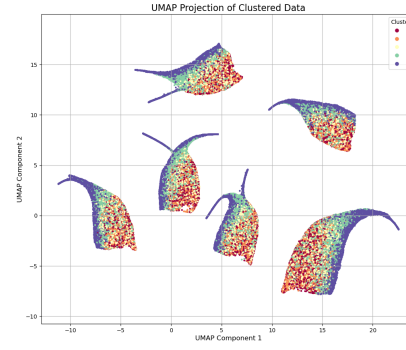
space, confirming that our methodology successfully captured the degradation trajectory.

Table 2: Interpretation of Degradation Clusters

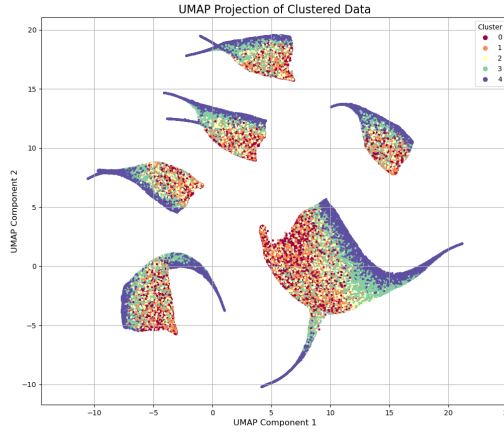
Cluster	Degradation Stage
0	Normal
1	Slightly Degraded
2	Moderately Degraded
3	Critical
4	Failure



FD001 $\cup$ FD003



FD002 $\cup$ FD004



FD001 $\cup$ FD002 $\cup$ FD003 $\cup$ FD004

Table 3: UMAP projections of the CMAPSS engine datasets, colored by weighted k-means clusters (labels 0–4), illustrating both operating-condition separability and degradation-stage progression. In (a), single-condition data (FD001 $\cup$ FD003) form one continuous manifold. In (b), six distinct manifolds correspond to the six operating conditions in FD002 $\cup$ FD004. In (c), the combined dataset preserves these six condition-specific manifolds. Within each manifold, cluster labels progress from **4** (end life) at the periphery through **3** $\rightarrow$ **2** $\rightarrow$ **1** $\rightarrow$ **0** toward the interior (less degradation), demonstrating that weighted k-means reliably captures the engine’s remaining useful life across varying operational settings.

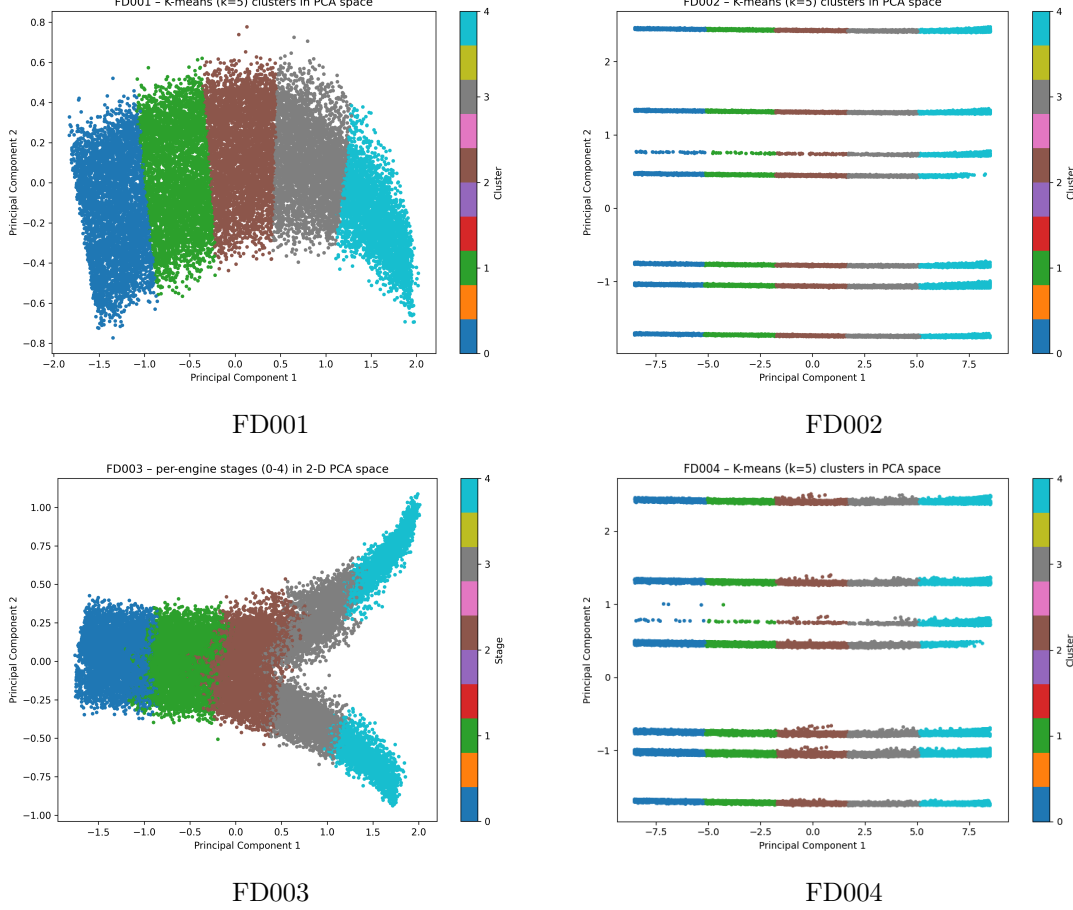


Table 4: PCA-based cluster visualizations for individual FD datasets.

4.2 Phase 2: Classification

After successfully generating pseudo-labels for degradation stages using weighted K-Means clustering in Phase 1, we transitioned the problem into a supervised classification task. The cluster labels obtained previously were treated as ground truth classes, enabling us to train classification models that could automatically predict the degradation stage of an engine at any given cycle based on its sensor readings and operational parameters.

To accomplish this, we experimented with a set of robust and widely used classifiers - Random Forest[1], Logistic Regression [4], and XGBoost [3] - each offering a different balance between interpretability, performance, and complexity. These models were trained independently on the labeled datasets generated for each of the four FD subsets (FD001 to FD004). This per-dataset approach ensured that model training respected the operational uniqueness of each engine type and fault condition, thereby maximizing classification relevance and accuracy within each domain. We also applied the same classification and evaluation procedure on the fully combined dataset ($FD001 \cup FD002 \cup FD003 \cup FD004$) as well as on the merged subsets $FD001 \cup FD003$ and $FD002 \cup FD004$ to assess model generalization across aggregated data configurations.

Each model was trained using the same input features that were used during clustering. To evaluate performance, we used a comprehensive suite of classification metrics:

- Accuracy to assess overall correctness.
- Precision and Recall to measure class-specific prediction quality.
- F1-Score as a harmonic mean of precision and recall, offering a balanced metric especially useful in imbalanced class distributions.

In addition to these numerical metrics, we also generated confusion matrices for each classifier-dataset pair to visualize misclassification patterns across degradation stages. These matrices provided insight into which stages were more prone to confusion, helping us understand model behavior beyond aggregate scores.

Among the models, **Random Forest** [1] consistently demonstrated strong performance, likely due to its ensemble nature and robustness to noisy or redundant features. XGBoost also performed competitively, particularly in datasets with more complex degradation behavior, while Logistic Regression, though more interpretable, sometimes underperformed in capturing nonlinear decision boundaries.

Overall, this phase allowed us to develop reliable classifiers capable of assigning degradation stages to incoming engine data, laying the foundation for more advanced predictive tasks such as estimating Remaining Useful Life in subsequent phases.

Confusion Matrix Comparison Across Models

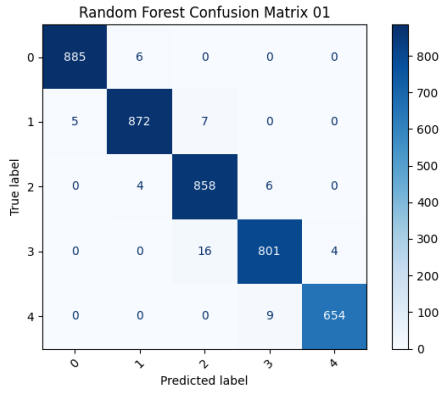
To better understand how each classifier performed in identifying the degradation stages, we present confusion matrices for all four FD datasets (FD001 to FD004) across the three models: Random Forest, XGBoost, and Logistic Regression. These matrices visualize the number of correctly and incorrectly classified instances per stage, highlighting performance trends across different fault modes and operational conditions.

Table 5: Evaluation Metrics (Accuracy, F1-Score, Precision, Recall) Across FD Datasets for All Classifiers

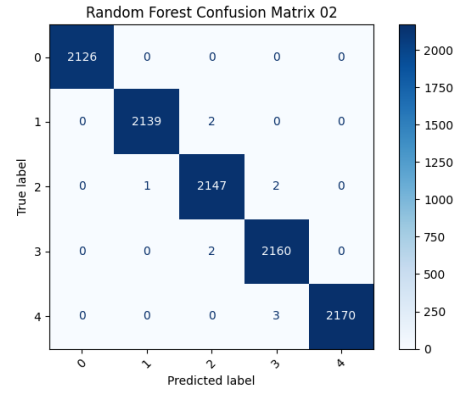
Dataset	Random Forest				XGBoost				Logistic Regression			
	Acc.	F1	Prec.	Rec.	Acc.	F1	Prec.	Rec.	Acc.	F1	Prec.	Rec.
FD001	0.986	0.986	0.986	0.986	0.987	0.987	0.987	0.987	0.984	0.984	0.984	0.984
FD002	0.999	0.999	0.999	0.999	0.996	0.996	0.996	0.996	0.993	0.993	0.993	0.993
FD003	0.932	0.933	0.934	0.933	0.933	0.934	0.934	0.934	0.933	0.934	0.934	0.934
FD004	0.999	0.999	0.999	0.999	0.996	0.996	0.996	0.996	0.996	0.996	0.996	0.996
FD001 $\cup$ FD003	0.996	0.996	0.996	0.996	0.995	0.995	0.995	0.995	0.987	0.987	0.987	0.987
FD002 $\cup$ FD004	0.999	0.999	0.999	0.999	0.996	0.996	0.996	0.996	0.995	0.995	0.995	0.995
Combined All	1.000	1.000	1.000	1.000	0.996	0.996	0.996	0.996	0.997	0.997	0.997	0.997

Note: “Combined All” refers to $\text{FD001} \cup \text{FD002} \cup \text{FD003} \cup \text{FD004}$.

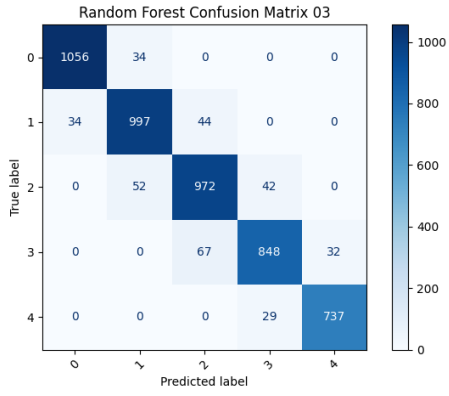
Random Forest Classifier – Confusion Matrices



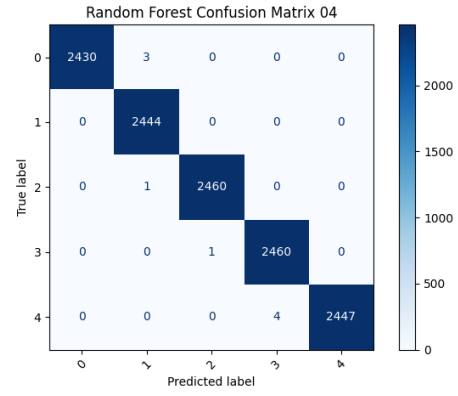
FD001



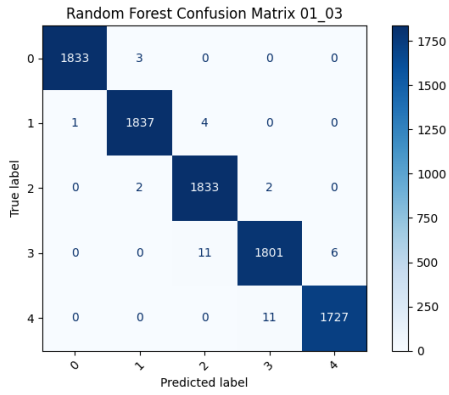
FD002



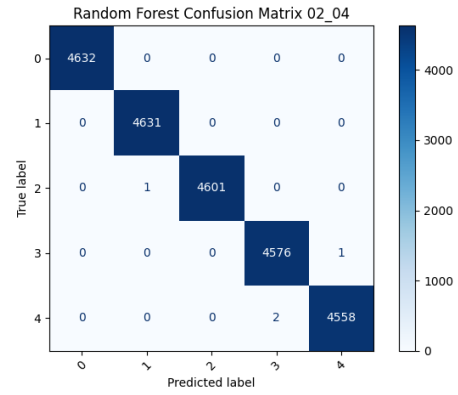
FD003



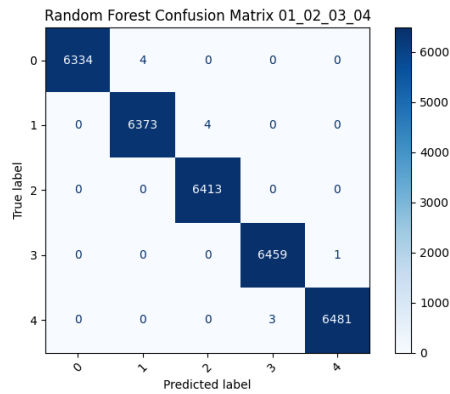
FD004



FD001 $\cup$ FD003



FD002 $\cup$ FD004



FD001 $\cup$ FD002 $\cup$ FD003 $\cup$ FD004

Figure 2: Confusion matrices of the Random Forest classifier on individual datasets (top two rows), pairwise combinations (third row), and the fully combined (bottom row).

4.3 Phase 3: Regression - Time-to-Next-Stage Prediction

Following the successful classification of degradation stages, the third and final phase of our hybrid pipeline focuses on predicting the Remaining Useful Life (RUL) in terms of cycles remaining until the engine transitions to the next degradation stage. Rather than predicting RUL until complete failure, we focused on a stage-wise RUL formulation, which is more interpretable and actionable for real-time maintenance planning.

To facilitate stage wise RUL prediction, we generated a new target column which represents the number of cycles remaining until the next degradation stage for each engine cycle. This was computed by identifying, for each data point, how many cycle steps remained before the engine transitioned into a higher degradation stage based on the clustered labels from Phase 1. For instance, if a data point was labeled as belonging to Stage 2 and Stage 3 commenced 40 cycles later for that engine, the associated rul true value was set to 40. This process was carefully repeated for each engine in all four FD subsets, allowing us to construct a meaningful, time-based regression target that aligned with the previously defined stage labels. Although this method assumes that stage transitions are temporally sequential and irreversible, it offers a practical and interpretable alternative to predicting RUL until absolute failure particularly useful for real time maintenance scheduling and localized decision-making.

Once these labels were constructed, the problem was framed as a supervised regression task. We used the same preprocessed features (sensor readings and operational settings) to train models that could predict the number of cycles remaining until the next stage transition.

We experimented with the following regression models:

- **Random Forest Regressor [1]** — a non-parametric ensemble method known for handling nonlinear relationships and interactions between features effectively.
- **Ridge Regression [6]** — a linear model with L2 regularization, useful for handling multicollinearity and producing stable coefficients.
- **Histogram-Based Gradient Boosting Regressor [8]** — a boosting ensemble that builds decision trees sequentially by fitting residuals of previous models, using histogram-based feature binning to speed up training and reduce memory usage. It often achieves superior accuracy on large datasets and can natively handle missing values.

Each model was trained and evaluated separately for the FD001–FD004 datasets to account for the distinct operating conditions and fault modes within each. This dataset-specific approach allowed the regressors to adapt to the varying degradation behaviors observed across engine types.

To assess model performance, we used standard regression evaluation metrics:

- **Root Mean Squared Error (RMSE)** — penalizes larger errors more heavily and gives insight into typical prediction deviations.
- **Mean Absolute Error (MAE)** — measures the average absolute difference between predicted and true RUL values.

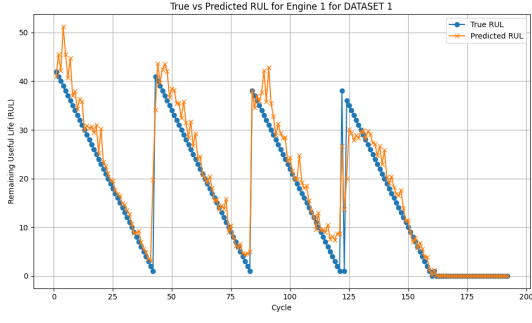
- **R² Score** — indicates the proportion of variance in the true RUL that is explained by the model.

Across experiments, both models demonstrated the ability to learn meaningful mappings from sensor readings to stage-wise RUL, with Random Forest Regressor generally outperforming Ridge Regression due to its flexibility in modeling nonlinearity. This phase added a crucial temporal dimension to our pipeline - enabling not just the recognition of the current degradation state, but also estimating how long the engine is likely to remain in that state before progressing to the next.

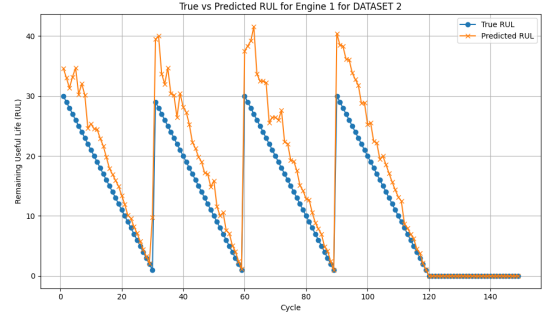
Table 6: Regression Model Evaluation Across FD001–FD004 Datasets

Dataset	Ridge			Random Forest			Hist. GB		
	MAE	RMSE	R ²	MAE	RMSE	R ²	MAE	RMSE	R ²
FD001	10.4164	13.1624	0.3026	2.9082	4.8765	0.9043	3.9969	6.1596	0.8473
FD002	10.2222	12.8128	0.2909	2.5711	4.2187	0.9231	3.2499	5.2963	0.8788
FD003	13.9500	18.3470	0.3234	5.8192	9.7269	0.8098	8.2768	12.9126	0.6649
FD004	13.0225	16.4504	0.2816	3.4147	5.6099	0.9165	4.2997	6.8434	0.8757
FD001+FD003	11.8380	15.4410	0.3400	2.9174	5.0226	0.9301	4.0674	6.7393	0.8742
FD002+FD004	11.8160	14.9490	0.2900	3.3680	5.4855	0.9044	3.9657	6.3849	0.8705
Combined All	11.8310	15.0820	0.2950	3.3584	5.5529	0.9045	3.9617	6.4834	0.8698

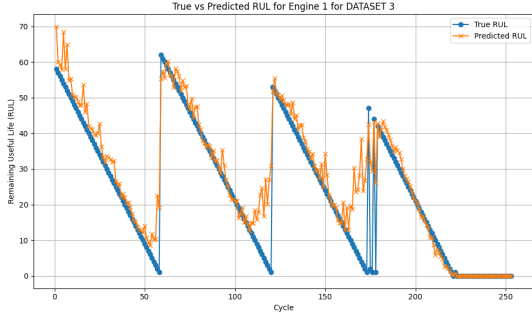
Note: “Combined All” refers to FD001 $\cup$ FD002 $\cup$ FD003 $\cup$ FD004.



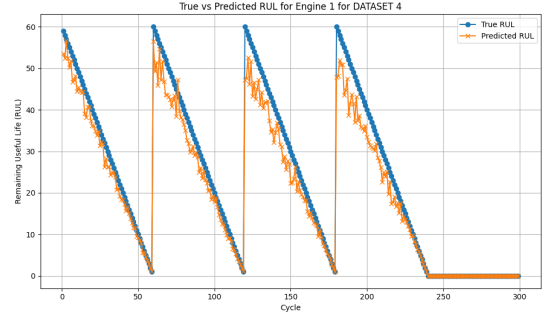
(a) FD001



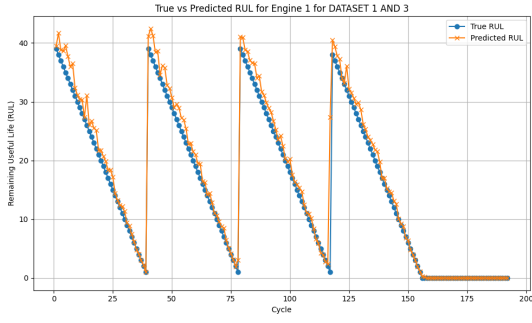
(b) FD002



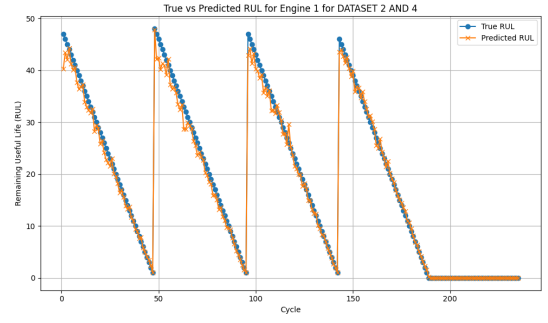
(c) FD003



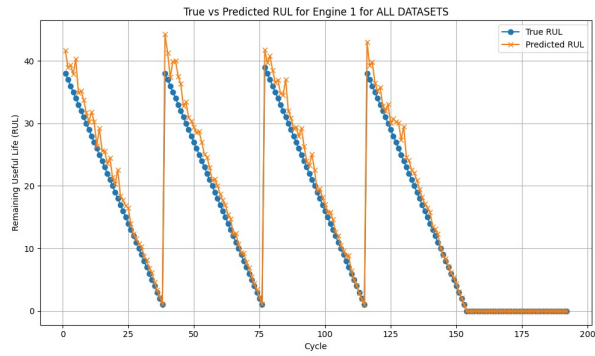
(d) FD004



(e) FD001 $\cup$ FD003



(f) FD002 $\cup$ FD004



(g) FD001 $\cup$ FD002 $\cup$ FD003 $\cup$ FD004

Figure 3: True vs. Predicted RUL for the Random Forest Regressor on (top two rows) individual FD datasets, (third row) the two merged subsets, and (bottom) the fully combined dataset.

4.4 Phase 4: Risk Score Computation and Decision Logic

The final phase of our pipeline focused on integrating the outputs of both classification and regression models to compute a unified, interpretable **Risk Score** for each engine cycle. This score reflects not just the current severity of degradation, but also how soon the engine is expected to reach failure. The motivation was to create a data-driven metric that could support real-time, threshold-based maintenance decisions.

Rather than relying solely on predicted degradation stages or raw RUL values, we designed the Risk Score to combine two key components:

- The failure probability, which quantifies the likelihood that an engine is in the failure stage (Stage 4),
- The estimated time left until the engine reaches that failure stage.

The intuition is simple: a unit is considered riskier if it is both highly likely to be near failure and has very little time left before that failure is realized.

Failure Probability via Logistic Regression. To compute the probability of failure, we trained a Logistic Regression to predict if an engine belongs to stage 4. Unlike tree-based models, Logistic Regression provides calibrated probability scores by default through the softmax function. For each data point, we extracted the predicted probability that the engine belongs to Cluster 4 (Failure Stage). This probability served as the first component of the risk score.

Time to Stage 4 calculation. We calculated the time to Stage 4 for all of the engines. This served the second part of the risk score.

Risk Score Computation. The raw risk score was then computed as:

$$\text{Risk Score} = \text{Failure Probability} \times \text{Time to Stage 4} \quad (1)$$

Min-Max Normalization. Since raw scores could vary significantly across datasets and engines, we normalized the scores using Min-Max Scaling to map them onto a $[0, 1]$ range. This allowed consistent interpretation and thresholding. The normalized score was computed as:

$$\text{Normalized Score} = \frac{\text{Raw Score} - \min(\text{scores})}{\max(\text{scores}) - \min(\text{scores})} \quad (2)$$

Here, the minimum and maximum raw scores were computed across the risk scores of the respective dataset.

Threshold Determination via Risk Score Visualization. After computing and normalizing the risk scores across all four FD subsets, we sought to determine a practical threshold for triggering maintenance alerts. Instead of arbitrarily choosing a cutoff value, we adopted a data-driven visual analysis approach. For each dataset, we plotted a line graph of the normalized risk score against the engine cycle index. These plots provided

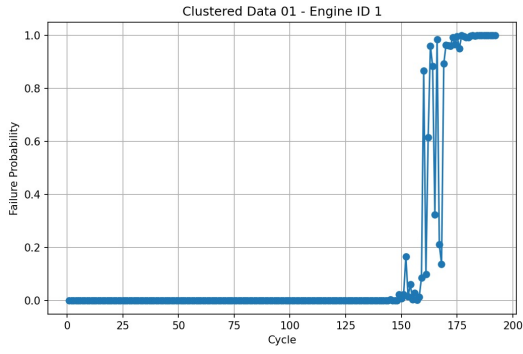
a clear temporal trajectory of risk for each engine, showing how the score evolved over time as degradation progressed.

Upon visual inspection of these graphs across all FD001 to FD004 datasets, we observed a consistent and interpretable pattern: whenever the normalized risk score crossed the value of 0.6, the corresponding data points were either already in Cluster 4 (Failure Stage) or transitioned into it shortly thereafter. This trend was stable across engines and datasets, indicating that 0.6 served as a reliable and generalizable threshold for identifying engines that were at imminent risk of failure.

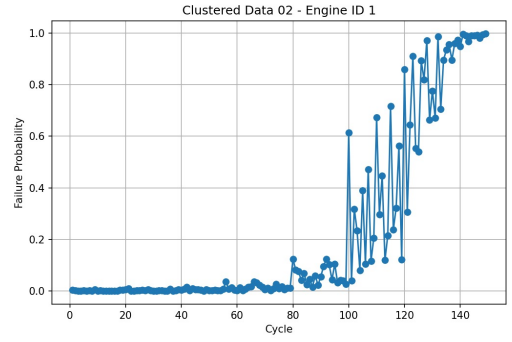
This observation allowed us to formally set 0.6 as the operational threshold in our decision logic. Any engine instance with a normalized risk score exceeding 0.6 was flagged as high-risk, prompting the system to issue a proactive maintenance alert. This threshold-based mechanism integrates smoothly with real-time monitoring systems and enables timely intervention, reducing the likelihood of catastrophic failure or unplanned downtime.

By grounding the threshold selection in visual evidence from actual model outputs, we ensured that our decision logic was not only technically sound but also empirically validated through cross-dataset consistency.

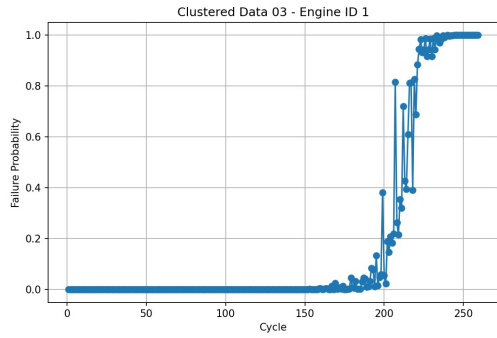
This phase operationalized the predictive capabilities of our classification and regression models by combining them into a single, interpretable metric. The use of Logistic Regression enabled reliable probability estimation, while the regression model contributed a temporal dimension. Together, the Risk Score not only captured the current risk of failure but also estimated how soon it might happen - fulfilling the goal of informed, timely, and automated maintenance planning. We performed the same threshold-crossing analysis on the merged subsets ($FD001 \cup FD003$ and $FD002 \cup FD004$) as well as on the fully combined dataset ($FD001 \cup FD002 \cup FD003 \cup FD004$) to confirm that the 0.6 cutoff remained robust across aggregated configurations.



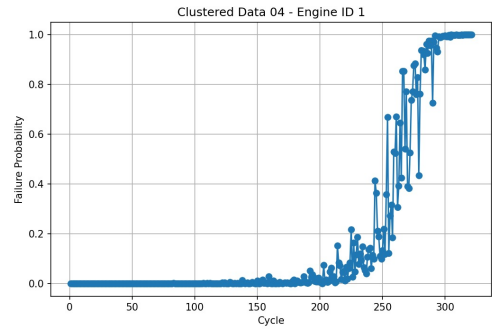
(a) FD001



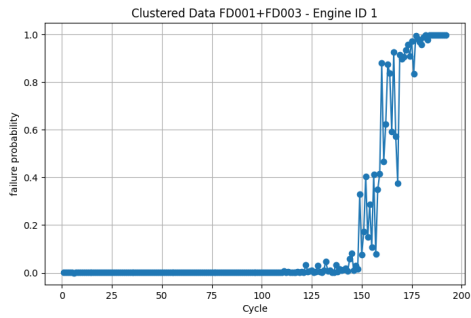
(b) FD002



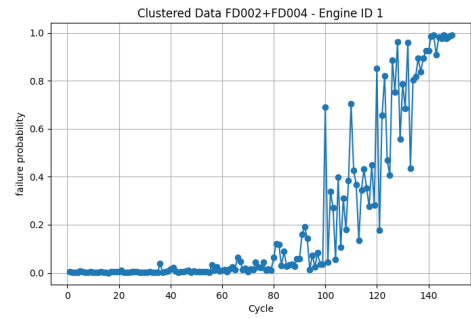
(c) FD003



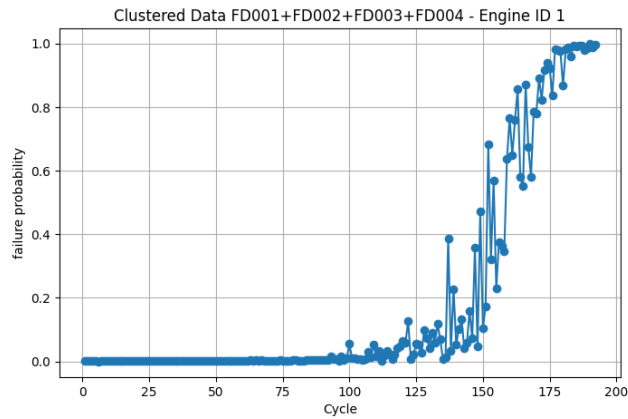
(d) FD004



(e) $FD001 \cup FD003$



(f) $FD002 \cup FD004$



(g) $FD001 \cup FD002 \cup FD003 \cup FD004$

Figure 4: Cycle vs. Normalized Risk Score for the Logistic Regression failure-probability plots on (top two rows) individual FD datasets, (third row) the two merged subsets, and (bottom) the fully combined dataset. 14

5 Conclusion

In this project, we presented a comprehensive and modular pipeline for hybrid predictive maintenance using the NASA CMAPSS dataset. By moving beyond traditional binary failure models, we addressed the need for more interpretable and actionable maintenance strategies that recognize the gradual nature of mechanical degradation.

We began by applying unsupervised clustering techniques – including a custom weighted K-Means algorithm – to derive five distinct degradation stages, which allowed us to convert unlabeled sensor data into a structured classification problem. This approach preserved degradation progression across engine life while enhancing stage separability.

In the second phase, we trained multiple classification models including Random Forest, Logistic Regression, and XGBoost to predict the degradation stage based on operational features. Random Forest and XGBoost yielded high performance, while Logistic Regression provided essential probability estimates used in later stages.

The third phase focused on regression-based estimation of Remaining Useful Life (RUL) until the next degradation stage. We generated custom target labels and trained models like Random Forest Regressor and Ridge Regression to forecast time until stage transition. This intermediate RUL estimation provided a temporal context to degradation, essential for prioritizing actions.

Finally, we introduced a composite Risk Score that combines the classifier’s failure probability (from Logistic Regression) and the calculated time-to-failure. We normalized this score using Min-Max scaling and analyzed its trajectory across engine cycles. Through visualization and manual inspection, we found that a threshold of 0.6 effectively distinguished engines approaching failure. This allowed us to implement a real-time decision logic that proactively flags high-risk engines and supports timely maintenance interventions.

Overall, our framework demonstrated how combining unsupervised learning, supervised modeling, and risk analytics can produce a robust, scalable, and interpretable system for predictive maintenance. The modularity of our design ensures adaptability to other industrial datasets and evolving equipment conditions, paving the way for intelligent asset management systems that are both data-driven and domain-aware.

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